Hermenegildo Mashele

hrdluis4@gmail.com | +27 67 837 7552 | Mpumalanga, South Africa

PROFILE

Computer Science Honours student specialising in Artificial Intelligence, with practical experience in deep learning, world models, computer vision, optimisation and software development. Experienced in designing reproducible machine-learning experiments, implementing models in Python, PyTorch and C++, and evaluating systems using quantitative and visual analysis.

TECHNICAL SKILLS

Programming	Python, C++, GDScript, Haskell, Assembly
AI / ML	PyTorch, torchvision, deep learning, self-supervised learning, contrastive learning, representation learning, reinforcement learning, world models, ResNet-18, ResNet-50
Evaluation	MSE, MAE, PSNR, SSIM, linear evaluation, PCA, t-SNE, experiment reproducibility, hyperparameter analysis
Optimisation / Search	Genetic Algorithms, evolutionary optimisation, HNSW, LSH, nearest-neighbour search, A*
Graphics / Games	Godot, OpenGL, Vulkan, SDL, procedural content generation, Wave Function Collapse, FastNoiseLite
Mathematics	Linear algebra, calculus, probability, abstract algebra, complex analysis

EDUCATION

BSc Honours in Computer Science (Artificial Intelligence) *In progress*
University of Johannesburg
Relevant modules: Artificial Intelligence, World Models, Advanced AI, Optimisation, Big Data Analysis, Computer Graphics, Functional Programming.

BSc in Computer Science and Mathematics *University of Johannesburg*

EXPERIENCE

Tutor *Done*
Academy of Computer Science and Software Engineering, University of Johannesburg

- Support undergraduate students with programming, algorithms, practical exercises, assignments and core computer science concepts.
- Explain technical concepts, assist with debugging, and help students develop structured problem-solving approaches.

SELECTED PROJECTS

Improving Long-Term Simulation Accuracy through Self-Correcting World Models *Honours Research Project, Completed*

- Built an Encoder–GRU–Decoder world model in Python and PyTorch for long-horizon prediction in Atari Breakout using 10,000 preprocessed 64×64 grayscale frames.
- Compared open-loop, fixed-interval and adaptive observation-assisted correction using the same frozen model across five 100-frame evaluation windows.
- Adaptive correction reduced mean MSE by approximately 37% compared with the open-loop baseline, while fixed correction reduced mean MSE by approximately 32%.
- Identified that low whole-frame error can hide incorrect small-object dynamics such as ball position, ghosting and motion drift.

Data Augmentation for Contrastive Representation Learning *Honours Advanced AI Research Project*

- Developed a reproducible SimCLR-style self-supervised learning pipeline in PyTorch using a CIFAR-adapted ResNet-18 and NT-Xent-style contrastive loss.
- Compared random cropping, rotation, colour jitter and Gaussian noise across 12 controlled experiments over three random seeds.
- Evaluated frozen representations using linear classification and t-SNE; random cropping achieved 47.74±0.85% mean classification accuracy.

Scalable Open-Set Wildlife Re-Identification *Big Data Analysis, Group Project*

- Built an image-retrieval pipeline using PyTorch and ResNet-50 embeddings to identify individual animals and detect unknown identities in a dataset of more than 140,000 images.
- Benchmarked Exact Search, LSH and HNSW for accuracy, latency and memory use; HNSW achieved approximately 108× faster retrieval than Exact Search with comparable Top-1 accuracy.

NEXORA – Procedurally Generated Connected Settlement *Honours Computer Graphics / Game Development Project*

- Developed a top-down 2D game in Godot using Wave Function Collapse to generate road networks from North, East, South and West connection rules.
- Added seeded generation, critical-system anchors, flood-fill connectivity validation, procedural terrain, rule-based building placement, NPCs, wildlife and system-activation objectives.

Artificial Life Food-Seeking Agents Using a Genetic Algorithm *Honours Optimisation Project*

- Developed a C++ Artificial Life simulation in which a Genetic Algorithm evolves food-seeking agents with speed and vision genes.
- Implemented fitness design, energy modelling, truncation selection, crossover, mutation and elitist replacement, then evaluated mutation rates and population sizes using reproducible experiments.

ADDITIONAL PROJECTS

- **Reinforcement Learning Agent:** Implemented and trained an agent while exploring reward shaping and exploration–exploitation trade-offs.
- **A* Pathfinder:** Implemented A* search in a purely functional programming style.
- **OpenGL / SDL Games:** Built small games covering rendering pipelines, input handling, game loops and basic physics.
- **Vulkan vs OpenGL Experiments:** Built graphics demos to compare low-level rendering performance on low-end systems.
- **Blockchain, OS Concepts and Assembly:** Completed practical coursework in hashing, consensus, chain integrity, operating systems and low-level programming.

RESEARCH AND ENGINEERING STRENGTHS

- Reproducible experiment design, controlled comparisons, random seed management and checkpointing.
- Model evaluation using both quantitative metrics and qualitative visual analysis.
- Comfortable moving between high-level machine learning work and lower-level implementation in C++ and systems-oriented coursework.
- Strong interest in AI research, computer vision, predictive models, optimisation and intelligent software systems.